

Levelling up

Everything the course assumes you already have: powers, expanding, factoring, fractions, equations, inequalities and systems.

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- 0.1 Powers and roots
 - 0.2 Working with algebraic expressions
 - 0.3 Factoring
 - 0.4 Fractions
 - 0.5 Linear equations
 - 0.6 Quadratic equations
 - 0.7 Turning a situation into an equation
 - 0.8 Inequalities and percentages
 - 0.9 Systems of linear equations
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0.1 Powers and roots

LEARN IT ON KHAN ACADEMY

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- ▶ **Video** The five rules · Exponent properties review
 - ▶ **Video** Simplifying square roots
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PRACTISE IT ON KHAN ACADEMY

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- ✍ **Exercise** The rules together · Properties of exponents challenge (integer exponents)
 - ✍ **Exercise** Simplify square roots
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- 1 **Same base, multiplied: add the exponents.**

$$a^m \cdot a^n = a^{m+n}$$

- 2 Same base, divided: subtract them.

$$\frac{a^m}{a^n} = a^{m-n}$$

- 3 A power of a power: multiply them.

$$(a^m)^n = a^{mn}$$

- 4 A product inside a power: the power reaches every factor.

$$(ab)^n = a^n b^n$$

- 5 Zero and negative exponents are definitions, not tricks.

$$a^0 = 1 \quad (a \neq 0) \quad a^{-n} = \frac{1}{a^n}$$

- 6 A root is a fractional exponent, which is why the same five rules cover roots.

$$\sqrt[n]{a} = a^{1/n} \quad \text{the } n\text{-th root of } a = a^{1/n}$$

SIMPLIFY $\frac{(2x^3)^4}{4x^5}$

$$\frac{16x^{12}}{4x^5}$$

The fourth power reaches the 2 as well as the x^3 , so $2^4 = 16$ and x^{12} .

$$4x^7$$

Divide the numbers, subtract the exponents.

WATCH OUT

■ $(a + b)^2$ is not $a^2 + b^2$. A power does not reach across a plus.

■ a^{-n} is not negative. It is a reciprocal, so $2^{-3} = \frac{1}{8}$.

■ $\sqrt{a^2 + b^2}$ does not split into $a + b$. Test it at $a = b = 3$.

0.2 Working with algebraic expressions

LEARN IT ON KHAN ACADEMY

► **Video** Expanding, from the start · Multiplying binomials intro

► **Video** Two brackets · Multiplying binomials

≡ **Article** Worked through · Warmup: Multiplying binomials

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✍ **Exercise** Expanding · Multiply binomials

✍ **Exercise** The difference of two squares · Multiply difference of squares

TERM

What the + and – signs separate. In $3x^2 - 5x + 2$ there are three terms.

FACTOR

What multiplies the whole of something. Factors cancel; terms never do.

LIKE TERMS

Same letter, same power. $3x$ and $5x$ are alike; $3x$ and $3x^2$ are not.

- 1 **Distribute across every term in the bracket.**

$$a(b + c) = ab + ac$$

- 2 **Two brackets: every term of the first against every term of the second.**

$$(a + b)(c + d) = ac + ad + bc + bd$$

- 3 **Two patterns are worth knowing by sight.**

$$(a + b)(a - b) = a^2 - b^2 \quad (a \pm b)^2 = a^2 \pm 2ab + b^2$$

- 4 A minus in front of a bracket reaches every term inside it.

$$-(x - 4) = -x + 4$$

- 5 Collect like terms last. Nothing else combines.

EXPAND $(2x - 5)(3x + 4)$

$$6x^2 + 8x - 15x - 20$$

Each term of the first bracket against each term of the second.

$$6x^2 - 7x - 20$$

Collect the two x terms.

WATCH OUT

• $2(x + 3)$ is $2x + 6$, not $2x + 3$. The 2 reaches both terms.

• $(q + 3)^2$ is $q^2 + 6q + 9$. The middle term is the one people drop.

• $3x$ and $3x^2$ do not combine. Different power, different term.

0.3 Factoring

LEARN IT ON KHAN ACADEMY

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- ▶ **Video** What factoring is · Introduction to factoring
 - ▶ **Video** Factoring a quadratic · Factoring quadratics intro
 - ▶ **Video** The difference of two squares · Factoring quadratics with difference of squares
 - ≡ **Article** The method, written out · Factoring simple quadratics review
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PRACTISE IT ON KHAN ACADEMY

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- ✍ **Exercise** Common factor first · GCF factoring introduction
 - ✍ **Exercise** Factoring a quadratic · Factoring quadratics intro
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- 1 Take out the common factor first, every time.

$$6x^2 + 9x = 3x(2x + 3)$$

- 2 For $x^2 + bx + c$, find two numbers that multiply to c and add to b .

$$x^2 - 13x + 36 = (x - 4)(x - 9)$$

- 3 Difference of two squares.

$$a^2 - b^2 = (a - b)(a + b)$$

- 4 Perfect square.

$$a^2 \pm 2ab + b^2 = (a \pm b)^2$$

- 5 **When the leading coefficient is not 1, split the middle term and group.** Find two numbers multiplying to ac and adding to b .
- 6 **Check by expanding.** It costs ten seconds and catches every sign error.

FACTOR $3x^2 + 11x - 4$

$$ac = 3 \times (-4) = -12$$

Two numbers multiplying to -12 and adding to 11 : 12 and -1 .

$$3x^2 + 12x - x - 4$$

Split the middle term using those two numbers.

$$3x(x + 4) - 1(x + 4)$$

Group in pairs and take each common factor out.

$$(3x - 1)(x + 4)$$

The bracket $x + 4$ is now itself the common factor.

WATCH OUT

- $x^2 + 9$ does not factor over the real numbers. A sum of two squares never does.
- Missing the common factor first makes every later step harder and sometimes impossible.
- $4x^2 - 16$ is $4(x - 2)(x + 2)$. Completely means completely.

0.4 Fractions

LEARN IT ON KHAN ACADEMY

► **Video** Dividing fractions by fractions

PRACTISE IT ON KHAN ACADEMY

 **Exercise** Adding and subtracting · Add and subtract fractions

 **Exercise** Dividing fractions

- 1 **Multiply straight across. No preparation needed.**

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

- 2 **Divide by turning the second one over.**

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

- 3 **Only adding and subtracting need a common denominator.** You can add halves to halves, not halves to thirds.

- 4 **Cancel factors, never terms.** A factor multiplies the whole numerator or the whole denominator.

- 5 **Keep the minus sign in front.**

$$\frac{-a}{b} = \frac{a}{-b} = -\frac{a}{b}$$

SIMPLIFY $\frac{x^2 - 25}{x + 5}$

$$\frac{(x - 5)(x + 5)}{x + 5}$$

Factor first. Only now is $x + 5$ a factor rather than a term.

$$x - 5$$

Valid for $x \neq -5$, since the original was undefined there.

WATCH OUT

■ $\frac{x + 5}{x}$ is not 5. The x on top is a term, so it does not cancel.

■ Test any cancellation with a number. At $x = 1$, $\frac{1 + 5}{1} = 6$, not 5.

■ Adding numerators and denominators separately is the single most expensive habit on this list.

0.5 Linear equations

LEARN IT ON KHAN ACADEMY

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- ▶ **Video** Unknown on both sides · Linear equations with variables on both sides
 - ▶ **Video** With brackets · Linear equations with parentheses
 - ≡ **Article** The method, written out · Multi-step equations review
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PRACTISE IT ON KHAN ACADEMY

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- ✍ **Exercise** Unknown on both sides · Equations with variables on both sides
 - ✍ **Exercise** Unknown in a denominator · Rational equations intro
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- 1 **Whatever you do, do it to both sides.** That is the only rule; the rest is order.
- 2 **Clear brackets first.** $-(x - 4) = -x + 4$.
- 3 **Then clear fractions by multiplying every term by the common denominator.** Every term, including the ones on the right.
- 4 **Collect the unknown on one side and the numbers on the other.**
- 5 **Divide by the coefficient of the unknown.**
- 6 **If the unknown sits in a denominator, multiply through by it, then check.** A value that makes any original denominator zero is not a solution.

SOLVE $\frac{2x + 1}{3} - \frac{x - 4}{2} = 1$

$$2(2x + 1) - 3(x - 4) = 6$$

Multiply every term by 6, the common denominator.

$$4x + 2 - 3x + 12 = 6$$

Expand. The minus reaches both terms of the second bracket.

$$x + 14 = 6$$

Collect.

$$x = -8$$

Check: $\frac{-15}{3} - \frac{-12}{2} = -5 + 6 = 1.$

WATCH OUT

- Multiplying by the common denominator and forgetting the right-hand side.
- $-3(x - 4)$ is $-3x + 12$. Both signs change.
- With the unknown in a denominator, an answer that makes it zero must be thrown away.

0.6 Quadratic equations

LEARN IT ON KHAN ACADEMY

- ▶ **Video** Solving quadratics by factoring
 - ▶ **Video** The formula · The quadratic formula
 - ≡ **Article** The method, written out · Solving quadratics by factoring review
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PRACTISE IT ON KHAN ACADEMY

- ✍ **Exercise** Quadratics by factoring
 - ✍ **Exercise** Using the formula · Quadratic formula
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- 1 Get everything to one side.

$$ax^2 + bx + c = 0, \quad a \neq 0$$

- 2 Read off a, b and c with their signs. The signs are where the marks go.
- 3 Try factoring. If it factors, it is faster. A product is zero only when one of its factors is zero.
- 4 Otherwise use the formula.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- 5 The discriminant answers how many roots before you find them.

$$\Delta = b^2 - 4ac \quad \Delta > 0: \text{ two} \quad \Delta = 0: \text{ one} \quad \Delta < 0: \text{ none in } \mathbb{R}$$

SOLVE $2X^2 - 7X + 3 = 0$

$a = 2, \quad b = -7, \quad c = 3$

Read them off with their signs.

$\Delta = 49 - 24 = 25$

Positive, so there are two roots, and 25 is a square so they are tidy.

$x = \frac{7 \pm 5}{4}$

Put it in the formula.

$x = 3 \quad \text{or} \quad x = \frac{1}{2}$

WATCH OUT

- **$b = -7$ gives $b^2 = 49$, positive. Squaring kills the sign.**
- **$(x - 2)(x + 5) = 4$ tells you nothing. Only a product equal to *zero* splits.**
- **A negative Δ means no real solution, which in a business question usually means the target is never reached.**

0.7 Turning a situation into an equation

LEARN IT ON KHAN ACADEMY

► **Video** A situation into an equation · Linear equation word problems

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✍ **Exercise** Writing the equation · Writing linear equations word problems

✍ **Exercise** Writing and solving · Linear equations word problems

✍ **Exercise** Two-step problems · Two-step equations word problems

- 1 **Name the unknown, with its unit.** Write it down. "Let s = the number of soups sold."
- 2 **Write every other quantity in terms of it.** If 80 items sold and s are soups, the rest is $80 - s$.
- 3 **Find the sentence that says two things are equal. That is the equation.**
- 4 **Solve it.** Now it is only algebra.
- 5 **Answer in words, with the unit, and check it against the situation.** A negative count or a price of 0.003 means the setup was wrong, not the arithmetic.

A STALL SELLS ROLLS AT 4.50 AND SOUPS AT 7.00. IT SOLD 80 ITEMS AND TOOK 440.00. HOW MANY SOUPS?

Let s = soups, so rolls = $80 - s$

One unknown is enough; everything else follows from it.

$$7s + 4.5(80 - s) = 440$$

The total takings is the sentence that says two things are equal.

$$2.5s + 360 = 440$$

Expand and collect.

$$s = 32$$

32 soups and 48 rolls. Check: $32(7) + 48(4.5) = 224 + 216 = 440$.

WATCH OUT

- Starting to write algebra before defining the letter. Half the lost marks are here.
- Mixing units. If one price is in francs, all of them are.
- Answering with a bare number. The question asked how many soups.

0.8 Inequalities and percentages

LEARN IT ON KHAN ACADEMY

- ▶ **Video** Multi-step inequalities

 - ▶ **Video** Percentages · Percent problems

 - ≡ **Article** Double inequalities · Compound inequalities review
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PRACTISE IT ON KHAN ACADEMY

- ✍ **Exercise** Multi-step linear inequalities

 - ✍ **Exercise** Percentages · Percent word problems
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- 1 An inequality solves like an equation, with one exception.
- 2 Multiplying or dividing by a negative flips the sign.

$$-2x > 6 \quad \Rightarrow \quad x < -3$$

- 3 A double inequality: do the same thing to all three parts.

$$2 \leq 3x - 1 \leq 8 \quad \Rightarrow \quad 1 \leq x \leq 3$$

- 4 A percentage is a multiplier. That is the whole idea.

$$p\% \text{ of } A = \frac{p}{100} \cdot A$$

- 5 Up by $p\%$ multiplies; down by $p\%$ multiplies too.

$$\times \left(1 + \frac{p}{100}\right) \quad \times \left(1 - \frac{p}{100}\right)$$

- 6 Successive changes multiply. They never add.

A PRICE RISES 20%, THEN FALLS 20%. WHAT IS THE NET CHANGE?

×1.20 then ×0.80

Each change is a multiplier on what came before.

$$1.20 \times 0.80 = 0.96$$

Multiply the multipliers.

a 4% fall

Not zero. The fall is taken on the larger price.

WATCH OUT

- $-2x > 6$ gives $x < -3$. Dividing by the negative flips it.
- Up 20% then down 20% is not back where you started.
- A percentage point is not a percent. From 4% to 6% is two points and a 50% rise.

0.9 Systems of linear equations

LEARN IT ON KHAN ACADEMY

► **Video** By substitution · Solving systems of equations with substitution

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✍ **Exercise** By substitution · Systems of equations with substitution

✍ **Exercise** By elimination · Systems of equations with elimination

- 1 **Two unknowns need two independent equations.** Independent means neither is a multiple of the other.
- 2 **Substitution: make one equation say $x = \text{something}$, then put that into the other.** The other one.
Putting it back into the equation you rearranged gives $0 = 0$.
- 3 **Elimination: scale one or both so a letter cancels when you add or subtract.**
- 4 **If the letters vanish and a false statement is left, there is no solution.** $0 = 5$. The lines are parallel.
- 5 **If the letters vanish and a true statement is left, there are infinitely many.** $0 = 0$. It was the same line twice.
- 6 **Check the pair in both equations.** One equation cannot catch an error.

SOLVE $3x + 2y = 19$ AND $x - y = 3$

$$x = y + 3$$

Rearrange the simpler of the two.

$$3(y + 3) + 2y = 19$$

Substitute into the *other* equation.

$$5y = 10, \quad y = 2$$

Expand and collect.

$$x = 5, \quad y = 2$$

Check both: $15 + 4 = 19$ and $5 - 2 = 3$.

WATCH OUT

- Substituting back into the equation you just rearranged. It always gives $0 = 0$.
- Subtracting equations and forgetting that the signs on the right change too.
- Checking in one equation only. A slip survives that.



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